

Deployment & Run Guide

Offline Docker bundle: install, run, demo, and troubleshoot the full SetuCredit stack without internet access — plus the borrower walkthrough for operators running a demo. Stack: React PWA (port 8080), FastAPI backend (8000), BRI scoring service (8001), Postgres, Redis.

Part A — Operator: deploy and run

1 What is in the bundle

The offline bundle is a single folder (**setucredit-deploy/**) that runs the whole system with no image pulls. Keep it together — the compose file refers to the image archives by name.

docker-compose.yml	Full 5-service stack (frontend, backend, scoring, postgres, redis). Pre-wired ports and in-network URLs; secrets read from .env with safe dev defaults.
.env.example	Copy to .env and edit. Every knob the stack reads (see §4).
images/*.tar.gz	Prebuilt Docker images, one archive each: backend, scoring, frontend, postgres:16-alpine, redis:7-alpine. Load with scripts/load-images.sh .
scripts/run.sh	Starts everything detached and waits until the backend reports healthy.
scripts/stop.sh	Stops the stack (data volumes kept; flag to wipe, see §8).
docs/*.pdf	This guide. Print it or keep it on the demo laptop.
SHA256SUMS	Integrity hashes for every file — verify before loading images.

2 Prerequisites

OS / CPU	64-bit Linux, macOS, or Windows + WSL2. Images are linux/amd64.
Docker	Docker Engine 24+ with Compose v2 (docker compose version).
RAM / disk	4 GB RAM free; ~4 GB free disk for images + DB volume. Bundled images total ~1.6 GB uncompressed.
Ports free	8080 (PWA), 8000 (API), 8001 (scoring), 5432 (Postgres), 6379 (Redis).
Network	None required after loading images — except SMS/ULI live modes and OTP delivery in a real demo.

3 Install and start (3 steps)

Step 1 — verify integrity (from the bundle root):

```
sha256sum -c SHA256SUMS
```

Step 2 — load the prebuilt images:

```
./scripts/load-images.sh
```

Step 3 — configure and start:

```
cp .env.example .env # edit secrets for your machine
./scripts/run.sh
```

run.sh starts all five containers detached, runs DB migrations automatically, and polls the backend health endpoint until it answers. Expect everything green within 1–2 minutes on a laptop.

4 Check it is up

PWA	http://localhost:8080/
API docs	http://localhost:8000/docs
Backend health	curl http://localhost:8000/healthz
Scoring health	curl http://localhost:8001/healthz

Healthy backend answers `{"status":"ok","postgres":"ok","redis":"ok"}`. If it answers **"degraded"** (HTTP 503), Postgres or Redis is not reachable — see §7. Scoring answers `{"status":"ok","model_version":"bri-heuristic-0.1.0"}`.

5 Configuration (.env)

Only `.env` needs editing; the compose file already wires service-to-service URLs. Unset optional hooks keep safe demo behaviour (simulated lender handoff, stub data, stub OTP).

ENV (default dev)	dev exposes the OTP on screen for demos (dev_otp). Set prod for any real user.
WEBHOOK_SECRET	HMAC key for lender webhooks. Default dev-webhook-secret — rotate in production.
PARTNER_WEBHOOK_URL (unset)	Partner NBFC endpoint. Unset = handoff recorded as "simulated" ; set it to send the real signed webhook.
ULI_MODE (stub)	Deterministic lending-history data. Set live with RBIH credentials for the real ULI adapter.
OTP_MODE (stub)	On-screen codes. Set sms with DLT sender/template for a real SMS provider.
OTP_MAX_ATTEMPTS (5)	Wrong-guess cap per code; resending resets it.

6 Two-minute demo script

Open <http://localhost:8080> (Chrome/Edge, volume up) and narrate the consent-first story:

1. **Language (10s)** — pick Hindi/Bengali/Tamil/Telugu/Marathi/English; UI and voice follow instantly.
2. **OTP (30s)** — enter the Aadhaar-linked mobile, read the on-screen code (dev mode), verify. Point out the 12-digit number is never asked or stored.
3. **Consent (20s)** — tick the sources; note it is time-bound, revocable, and enforced in code (pulls without consent get HTTP 403).
4. **Appraise (30s)** — parallel pull across land, electricity, bank, ULI → BRI score + suggested loan range.
5. **Result (30s)** — gauge with legends below, band note, biggest-lever tip, per-source breakdown, lender handoff status, Print / Save as PDF.

Score bands: **Strong 80–100** (larger loans) · **Good 60–79** (most partner lenders) · **Fair 40–59** (smaller loans; bills + activity improve it) · **Building 0–39** (build history, then check again). Loan-range suggestion appears from BRI 40 up.

7 Troubleshooting

Port already in use	docker compose ps ; free the port or stop the other stack. Compose error names the clash.
Backend "degraded" / 503	Postgres/Redis still starting or down: docker compose ps , then docker compose logs postgres redis . Data is intact; wait or restart.
Frontend blank / API unreachable	docker compose logs backend frontend . Check http://localhost:8000/healthz directly.
HTTP 403 on appraise	No valid consent covers the request — grant/re-grant consent in the app. This is the DPDP gate working as designed.
HTTP 409	Step run out of order (e.g. appraise before OTP). Follow the progress bar; or start a fresh session.

All sources failed / scoring down	docker compose logs scoring backend ; restart scoring. Partial-source results still score if the minimum set succeeded.
Fresh start	./scripts/stop.sh then ./scripts/run.sh . Nuclear: docker compose --profile app down -v wipes DB volumes.

8 Stop, update, back up

./scripts/stop.sh

Keeps the **pgdata** volume (audit ledger, consents, outcomes) so a restart resumes history. To wipe demo data between shows, use **docker compose --profile app down -v**. To update: replace the bundle folder, re-run load + run. Back up Postgres with **pg_dump** against localhost:5432 if the ledger matters.

9 Security notes (read before any real user)

- Bundle defaults are **demo credentials** (postgres password, webhook secret). Rotate them via **.env** before production.
- **ENV=dev** prints OTPs on screen — never use dev mode with real borrowers; set **ENV=prod** and **OTP_MODE=sms**.
- Pass-through by design: Postgres holds audit metadata, consent records, and appraisal outcomes only — never borrower records or features.
- Serve traffic over TLS (reverse proxy) and keep Redis/Postgres off the public internet.

Part B — Borrower walkthrough

For the operator to narrate — or print for the borrower. Five steps, about two minutes.

Step 1 — Language

Choose Hindi, Bengali, Tamil, Telugu, Marathi, or English on the first screen. Buttons, instructions, and the spoken voice switch immediately. Keep the volume up — every step is read aloud.

Step 2 — Mobile OTP (identity check)

Enter the mobile number linked to Aadhaar and tap Send. A 6-digit code arrives by SMS (on the demo screen it is also shown). Enter it to prove the session belongs to you.

- The 12-digit Aadhaar number is **never** asked for, spoken, or stored.
- No SMS? Check the Aadhaar-linked SIM is in the phone with signal, wait 60 seconds, then Resend (resending resets the 5-guess counter).
- Wrong number linked, or never linked? Check mAadhaar / UIDAI Verify-Mobile; linking or changes need an in-person visit to an Aadhaar Seva Kendra.

Step 3 — Consent (your permission slip)

Tick exactly the records you agree to share: land registry, electricity-bill payments, bank-account summary (Account Aggregator), lending history (RBI ULI). Nothing is read before you agree. Consent is time-bound and revocable — revoking stops the flow and aborts any in-flight pull. You may share fewer sources; the score still works but with lower confidence.

Step 4 — Automatic check (about 30 seconds)

The system pulls the consented records in parallel, turns them into a Borrower Readiness Index (BRI, 0–100), and prepares a signed package for the partner lender. You wait on a progress screen — nothing to fill in.

Step 5 — Result and next steps

A gauge shows the score with the band legends beneath it (Strong / Good / Fair / Building). Below: what the band means, the one action that would lift the score most ("biggest lever"), points earned per source, the partner-lender handoff status, and — from BRI 40 up — an indicative loan range (the lender sets the final amount). Use **Print / Save as PDF** to keep the report.

If the package could not reach the lender, start a new session and try again. The report's handoff line always says what happened ("simulated" on a demo machine means no lender endpoint is configured).

Your data and privacy, in one paragraph

Records are processed in memory for this request only and never stored — not even as derived features. Only the audit trail (what happened, when, how many records, consent reference), the consent itself, and the outcome (score, per-source ok/failed, handoff state) are kept, so the process can be audited. The lender receives appraisal ID, score, model version, and loan range — never raw records. SetuCredit never lends and never decides; the partner NBFC approves and pays out over UPI/IMPS.

Sources: README.md, docs/ARCHITECTURE.md, SetuCredit User Guide & Handbook, in-app help text. Bundle: setucredit-deploy/ (compose + prebuilt images + this guide). Endpoints: PWA :8080, API :8000, scoring :8001.